r/AskPhysics • 8mo ago
u/MerengueFinancial

Can worldlines, in principle, map... everything?

A thought I've been tossing around for decades (I've tried to work out the math, but honestly, it's a bit beyond me), but hear me out:

If we know all of the fundamental forces impacting a particle, and we know how that particle's behavior is impacted by each... then the particle's worldline is just a four dimensional curve which, I think in principle, would be an ideal way to map *all* of the forces on to a *single* fundamental structure. External forces determine the overall four-dimensional path of the worldline, and we already sort of map internal forces as oscillations, so strong, weak, electromagnetic... should, in principle, be mappable to a sort of series oscillations superimposed upon that worldline. And the same would hold true for any particle.

If that's doable, then it the geometry of the worldline should be describable with classical wave equations. Old math. But old math that encodes *everything*. Something like a reframing of covariant equations. You should, I would think, be able to get all of that into the equation of a single, very complex, four-dimensional helix.

If that seems reasonable, then looking at our worldline diagram, we draw our plane to represent the "now" moment, the intersection being the particle, and we'd see a very particular angle of intersection between the 'time sheet' and any given particle worldline. Treat that angle as a measure of something like 'drag'—just a transfer of energy—and we've got a possible mapping of mass—which could then be translated into a drag-induced curvature in the time surface.

If we look along the main long axis of the helix, we'd see a geometry resembling string theory modes. A quark could be seen as three braided worldlines. And quantization... might come out of the wrap rate of the helix as a sort of 'click rate'.

It doesn't have to have any physical reality, but just as a model, I'd think this view should be formalizable into mathematics in a fairly straightforward (if very complex) way, and might offer a path to unification.

Has this been tried before? Do other people think of worldlines in this way?

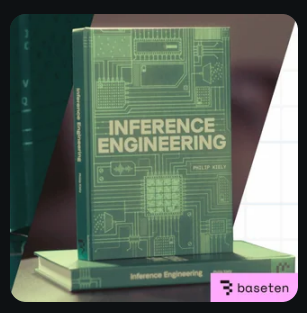
[For clarity... I'm not advancing any theory, I'm asking for existing physics, would this sort of constraint space be sufficient, in principle? Not asking whether It should or whether it's feasible to try, and certainly not suggesting any necessary cause-and-effect relationship between map and model, just... Can this work as a conceivable mathematical construct? Could a single worldline be viewed as the unique solution that satisfies all known laws under a given set of conditions, if those laws are isometrically mapped to worldline convolutions relative to each other?]

0 23

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unlikely_arrangement • 8mo ago

On a related note, I see references to the “block universe” in which time is represented geometrically. A single particle then has its entire existence described as a 4-dimensional curve. Add a second particle and you can describe both particles as a curve in either 7- or 8-dimensional space. You could also describe them as two curves in 4-space, but only if you are happy with a common time axis. If not, you can describe the entire universe over all time as a curve in a 10^{240} dimensional space. I only got interested in this because of the tricky concept of “the present moment” as experienced by me personally, and presumably you as well.

2

MerengueFinancial OP • 8mo ago

Yes, thinking about this idea led me to learn about Hamiltonians and Hilbert spaces, and I think it’s basically just a special case of both. I had been thinking in a block universe sort of way (which I think is justifiable only if treated as a stepping stone for ‘making the map’).

What I don’t understand is why dissipation would have to be an insoluble problem; or rather, why that insolubility would, as I think you’re suggesting, break the math—it’s just the limitation on how far you can propagate predictions in four space, and I think it would probably approximate the uncertainty principle.

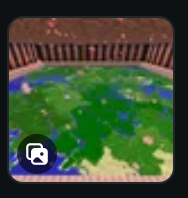
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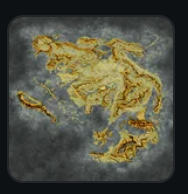
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I think this world map is complete, isn't it?



around orders of magnitude of dimensional space, you just choose the dimensional space you find most convenient to think in, and project the math down to one dimension, or to the minimal dimensions you can get away with. Which, I think would be our familiar 3+1, but treating the +1 as spacelike, just as in a block universe representation. It would just take all the convolutions up in 240-space or wherever, and move them down to a 1D line in 4-space. For no better reason than that it's easier to visualize. But either representation would be completely equivalent. Does that track?

1

unlikely_arrangement • 8mo ago

Yes, you bring up the real issue. Does either representation teach us anything? One book I read was very careful to observe that time moving through the block universe representation is not moving "in time" since it is time. It's certainly true that to me, the block universe could only represent all of history in retrospect, since it hasn't happened yet. However, if there were no sentient beings, and to be safe no life at all, the block universe doesn't require the concept of the passage of time. I know this sounds like BS, but I have put serious thought into this.

2

MerengueFinancial OP • 8mo ago

You're totally right, and I don't think it sounds like BS. Or, no more BS than any other abstract representation. It's just a choice of how to think about it, and the discipline to insist on putting *everything* into the *same* coherent framework out the gate, even if it's more convenient for a particular theory to think about it in another way. But *everything* is translatable into lines which means *all* physical theories should be, in principle, compatible with one another to the extent that they don't contain broken mathematics. Whether all events already exist or they're all unfolding dynamically, or they're something else and our perspective is what's shifting... that, to me, is philosophy. And I think all of those ontological perspectives boil down to interchangeable equivalences. imho

1

unlikely_arrangement • 8mo ago

Ah! I believe that the observed existence of a present moment, that one issue, is never actually addressed rigorously. But it is a clear observation, it is objectively real, and I believe that it is a completely valid question for physics to treat. Which means there should be testable theories, followed by tests. Not that I know what that would look like at the moment. And yes, any theory would have to be relativistically correct.

1

Continue this thread

triatticus • 8mo ago

If you draw a plane of simultaneity there is no "angle" with the time axis as you have a single point where that particle is at that particular time.

1

MerengueFinancial OP • 8mo ago

Within that plane, yes. But in the 4-space representation, you've got an angle of intersection between your hyperhelical worldline and the plane itself. If time is represented (and it is just an abstract representation, we can't know the 'reality') as a movement of that plane along its normal, then any two worldlines have a coordinate space in which to plot their trajectory relative to one another. Let your angle of intersection represent—well, whatever the geometry most comfortably accommodates... then plane and the normal, I think, would act as sort of a Hilbert space definition.

You could do it lots of different ways—I suggested a transformation of the plane, but you could just as well toss the 'time plane' and consider the angle between the helical slope and the long axis of the worldline itself. The choice of coordinate system is arbitrary—but by starting with a geometrically coherent primitive, or set of primitives, you give yourself a constrained set of mapping possibilities. Point being, coherent geometry offers a set of rules that seem like they might make mapping the big theories to one another a bit less daunting.

1

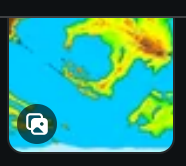
triatticus • 8mo ago

No, you're using concepts without understanding them. A "plane of simultaneity" is a slice of spacetime at a definite time coordinate. Here is no angle here to define, you define angles with tangents to the world line at a specific time and this only defines an angle in a given reference frame as this angle is frame dependant. Just look at the 2D case, the intersection of a worldline with a horizontal line (plane of simultaneity) is a point (event). If you move to 3D it's still a point and you can't assign a direction to a point the way you specifically are trying to.

r/worldbuilding • 5mo ago

The first maps of my world! Ask me anything or leave some...

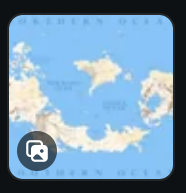
36 upvotes · 13 comments



r/mapmaking • 27d ago

Mostly finished world map

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r/mapmaking • 22d ago

Is this world geographically plausible

295 upvotes · 100 comments



r/rainworld • 5mo ago

My interpretation of the rain flat world map yay

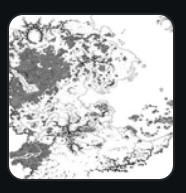
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r/mapmaking • 6mo ago

Worldmap finished (for now)

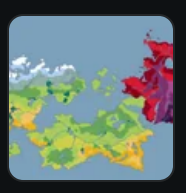
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r/worldbuilding • 17d ago

The geographic map of my world

33 upvotes · 14 comments



r/worldbuilding • 10mo ago

How do you map your world?

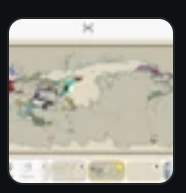
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r/mapmaking • 5mo ago

unfinished world map

90 upvotes · 8 comments



r/geography • 19d ago

A completely normal world map

103 upvotes · 43 comments



r/PossibleHistory • 8mo ago

The map of the world, but EVERY comment changes it. Day 0.

71 upvotes · 137 comments



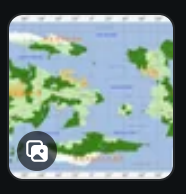
r/mathematics • 10mo ago

Question on Dimensions...conceptually what is a negative dimension?

13 upvotes · 51 comments

r/imaginarymaps • 4mo ago

World map of my world



...well, yes. If I understood, I'd be over in r/Physics ;P

But assigning a direction (or magnitude) to a point is just giving it a motion vector. Same principle as a point in a field equation. Isn't it?

The intersection is a point. 4-space is a coordinate system in which to plot that point's position over time. That position over time is a curve. And curves have tangents, which have angles relative to any arbitrary coordinate system you want to set up. The time plane (and any mathematical abstraction) is a convenient fiction. It might be simpler to consider the slope of a (hyper)helix. Then, it's an angle relative to its own long axis. Either way, if you treat a particle as a collection of superimposed helical oscillations on a worldline, you've got a set of primitives which have relationships to one another, one of which is an angle within your coordinate system, whether internal to the worldline, or measured against a representation of time (or any similar abstraction). Whether that angle is useful or meaningful within the math—let alone the universe at large—is another question.

1

MerengueFinancial OP • 8mo ago

Ohhhhhh... I see what you're saying.! That's right. A point does not have intrinsic properties apart from position. So: no machinery for interaction in an instantaneous moment. That's important. It means either everything falls apart... or no interaction without motion. Or. No points allowed

1

Internal-Sun-6476 • 8mo ago

There is no explanation for what a worldline is.

1

MerengueFinancial OP • 8mo ago

Just a representation of a particle's path over time. A mathematical curve in four dimensions.

1

MerengueFinancial OP • 8mo ago

Corrected, good call. Thank you.

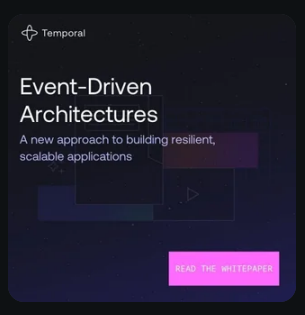
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NoInstruction75 • 8mo ago

This is a long answer but it essentially is to tell you "yes" to both your questions. It doesn't quite work the way you said but I think it is morally similar.

Strip away the question of worldlines and spacetime, and what you are describing sounds a lot like regular Hamiltonian mechanics. In Hamiltonian mechanics, a mechanical system follows trajectories in a very high dimensional phase space. The trajectories are curves in this high dimensional space and each curve represents the time evolution of a particular arrangement of particles at the initial time. In some cases, the trajectories turn out to be simple but in most cases they are intertwined throughout the phase space in a complicated way. The reason I bring it up, though, is that once I tell you where all the particles are and what their momenta are, I follow this trajectory and I can read off the position and momenta of all the particles at any other time. I think this is exactly what you asked for: there is a map that evolves a state forward in time and uniquely identifies a past state with a future state in an invertible way. These ideas extend nicely to spacetime and general relativity.

What are the issues? Once there is dissipation, you lose the nice one-to-one correspondence between the future and the past. Follow a pendulum long enough and it ends up hanging still and that means you cannot figure out how it started.

But maybe all particles evolve according to a Hamiltonian, which is reasonable because dissipation happens because you transferred momentum somewhere else, and so on. Then, well, isn't this what you are asking for?

2

Is it lazy to base the geography of your world on real world countries?

35 upvotes · 43 comments

r/mapmaking • 4mo ago

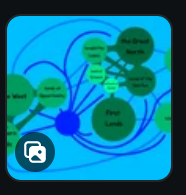
First worldmap for the setting I'm writing, looking for feedback



41 upvotes · 13 comments

r/worldbuilding • 5mo ago

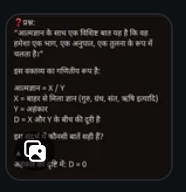
My world map!!



84 upvotes · 18 comments

r/AcharyaPrashant_AP • 8mo ago

Option 'D' should perhaps be something like this 😜... Gita...



7 upvotes · 1 comment

r/HypotheticalPhysics • 9mo ago

Here is a hypothesis: universe is made of variations

5 comments

r/goodworldbuilding • 2mo ago

What are the rules of your world? Why are they in place?

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MerengueFinancial OP • 8mo ago

Actually, that's exactly what I started out doing... years ago, thinking I was imagining string theory; later, I realized I'd misunderstood string theory and started thinking of this as "something else". But I couldn't get it out of my head, I just ended up trying to map everything in 4D in my head, and eventually I wondered... is this mathematically rich enough to do it all? Which is what I'm trying to ask here. But... it also happens that, if it is, then you look at it from one angle, and it does look kind of like string theory, look at it from another, it looks kind of like relativity. Just from looking at a standard worldline diagram, and the pieces—to me—seem to intuitively snap into place. Not as a theory, but as a way to put the theories we already know into one place so they can play together. Maybe. But I suspect there has to be a fatal flaw.... like a worldline diagram is inherently unsuitable to map any arbitrarily complex law of physics, or like mapping everything to a complex series of worldlines hits the same wall that everything else hits when trying to go from GR to QFT... otherwise, somebody would have done this already. Maybe someone has and it didn't work?

1

[deleted] • 8mo ago

Comment removed by moderator

4

MerengueFinancial OP • 8mo ago

I think I fundamentally agree with that. But I think in this outlook—or, likely, any outlook that forces *all* coupling to be calculated by transformations of a single mathematical primitive within a coherent geometric framework—what one would be putting together wouldn't be derived from, nor used to derive a theory of matter per se. The underlying theories would still just be approximations of observed behavior. You'd just be forcing the calculations that do the approximation into the same coherent universe of possibilities. Which would make them 'inter-coherent'. Or rather, would show that they have to be, either coherent with each other, or wrong, and perhaps in effect 'point out' where insoluble inconsistencies between the big theories actually live in the math in a way that suggests how adjustments might make them soluble. If that makes sense.

The idea, I see as just a sort of 'compatibility kludge'. Theories that apparently aren't compatible with one another... if they can all be translated into transformations of a simple line (or, presumably a sphere or loop or any other primitive), ought to be quietly compatible with one another. It's just a scaffolding that might help 'iron out the inconsistencies'.

1

[deleted] • 8mo ago

Comment removed by moderator

3

MerengueFinancial OP • 8mo ago

Yes, I think that would be true of the overall trajectory. But given that we already have models that work, then those, when applied to the single line, force constraints upon that line, and thereby upon each other (I *think*)...so, like a single worldline cannot in itself encode all the laws of physics... but integrating over those laws should give you all the possible worldlines—or something like that. I'm sure I'm not fully explaining what I mean.

When it comes to internal forces... at that level, all we have is models, right? In a broad sense, all we're ever doing at the fundamental particle scale is just looking at behaviors, and coming up with math that fits. That's my understanding anyway. So... we just 'do a math' on it and if the math works, great. The only thing I'm wondering is this: If we do this math and that math (that we know works) *in the same constrained one-dimensional space*... maybe arbitrarily choosing the mapping for, say, color charge... does that *force* a particular mapping of all the others? In principle, a line ought to be able to encode a system of any complexity—but when forced to do so for *all* known forces simultaneously... do they "snap together" in reality like they seem to in my head? If that makes sense.

1

MerengueFinancial OP • 8mo ago

Things like mass would have to be some similarly constrained property of the structure. So... maybe mass would just be encoded as variation in curvature rate. There are probably multiple ways to use a 4D curve as the basis for a model; I suggested intersection angle in my post, but at heart, my question isn't about any particular way of constructing such a mapping, rather whether such a mapping, however one chooses the particulars, should be possible at all.

1

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